Bo Yang

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Education

Columbia University — **Aug. 2025 – Present**
Ph.D. in Marketing | GPA: 3.8/4.0 — *New York*

University of Michigan, Ann Arbor — **Sep. 2023 – May 2025**
M.S. in Biostatistics | GPA: 4.0/4.0 — *Ann Arbor, MI*

University of Michigan, Ann Arbor — **Sep. 2020 – May 2023**
B.S. with High Honors. Data Science, Math, and Statistics | GPA: 3.94/4.0 — *Ann Arbor, MI*

Research Interests

- Multiagent reinforcement learning, Statistical Machine Learning, Deep Learning, Nonparametric analysis, Diffusion Model, Latent States Modeling, Bayesian Modeling

Publication

- Quantifying Uncertainty in Classification Performance: ROC Confidence Bands Using Conformal Prediction (Arxiv)
- Revisiting Classification Uncertainty Quantification: Comparison between Bootstrap and Conformal ROC Confidence Bands (Published in Statistica sinica)
- Mechanistic Models for Panel Data: Analysis of Ecological Experiments with Interacting Species using a Partially Observed Markov Process Framework (In revision by Annals of Applied Statistics)
- Conditional Prediction ROC Bands for Graph Classification (Accepted by *AISTATS* 2025)
- Conditional Uncertainty Quantification for Tensorized Topological Neural Networks (Submitted to *PAKDD* 2025)
- Statistical Hypothesis Testing for Efficient Black-Box Quantum Program Testing (Minor Revision in IEEE Access)
- Modeling Story Expectations to Understand Engagement: A Generative Framework Using LLMs (In revision by Marketing Science)
- Does eKare Outperform Standard of Care in Characterizing Wound Healing? Evidence from a DFUC Ancillary Study (Submitted to International Wound Journal)

Research Experiences

Statistics Division, Researcher, Umich — **Aug. 2022 – Present**
Mechanistic Modeling, Statistical Learning, Time Series Analysis — *with Prof. Edward Ionides*

- Integrate mechanistic models with the Panel Partially Observed Markov Process (PanelPOMP) framework for complex ecosystem analysis.
- Apply the Auto-Differential Iterated Filtering method to panel data analysis, advancing time series modeling.
- Submitted Paper: Mechanistic Models for Panel Data: Analysis of Ecological Experiments with Four Interacting Species (Submitted to AoAS)

Biostatistics Division, Researcher, Umich — **Sep. 2023 – May 2025**
Conformal Prediction, Statistical Learning, Package Development

- Developed innovative conformal prediction methodologies to quantify uncertainty in ROC curves and AUC, providing a novel alternative to bootstrap techniques in model uncertainty quantification.
- Addressed limitations of traditional bootstrap methods, enhancing uncertainty quantification in classification performance metrics.Validated the effectiveness and reliability of conformal prediction through theoretical analysis and extensive simulations under both i.i.d. and non-i.i.d. test data scenarios. Developing the *CPROC* R package for implementing conformal prediction in ROC analysis, facilitating practical applications.
- Preprint: Quantifying Uncertainty in Classification Performance: ROC Confidence Bands Using Conformal Prediction (To be submitted to *Annals of Statistics*)
- A Revisit on Classification Uncertainty Quantification: Comparison between Bootstrap and Conformal ROC Confidence Bands (Accpeted by *Statistica Sinica*)

Computer Science Division, Researcher, Umich **Mar. 2024 – May 2025**
Deep Learning, Computer Vision, Conformal Prediction

- Developed methods for graph classification by integrating conformal prediction and tensor-based neural networks, enhancing reliability and precision in graph-structured data analysis.
- Addressed statistical reliability concerns in Graph Neural Networks (GNNs) by introducing Conformalized Tensor-based Topological Neural Networks (CF-T2NN). Employed CF-T2NN to quantify uncertainty in non-exchangeable graph-structured data, reducing prediction set sizes in graph classification tasks. Validated the superiority of CF-T2NN over methods across 10 real-world datasets, setting new standards in graph-structured data analysis.
- Applied Conformal Prediction to construct ROC bands for robust uncertainty quantification of ROC curves under distributional shifts in testing data.
- Validated the superiority of CF-T2NN over methods across 10 real-world datasets, setting new standards in graph-structured data analysis.
- Conditional Prediction ROC Bands for Graph Classification (Accepted by *AISTATS* 2025)
- Conditional Uncertainty Quantification for Tensorized Topological Neural Networks (Submitted to *PAKDD* 2025)

Marketing Division, Research Assistant, Umich **Mar. 2024 – May 2025**
Bayesian Modeling, Latent States Modeling *with Profs. Fred Feinberg, Gwen Ahn, and Kee Yeun Lee*

- Developed advanced Bayesian models to predict donors' latent trajectories of donation likelihood and amount, optimizing fundraising strategies through personalized appeal scales.
- Enhanced computational efficiency by rewriting the model in NumPyro, achieving a 100x speed improvement.
- Analyzed donor behavior and evaluated model performance using parameter recovery plots, predictive analyses, and confidence intervals.
- Performed counterfactual analysis to measure the effects of scale manipulation and identify optimal appeal strategies for individual donors.

Marketing Division, Researcher, UNC **Jun. 2024 – Aug. 2025**
Causal Inference, Conformal Prediction, Machine Learning *with Prof. Longxiu Tian*

- Conducted conformal prediction analysis to enhance causal estimation of price promotions' effects on individual acquisition with quantified uncertainty.
- Applied XGBoost model and onformal prediction techniques to estimate individualized treatment effects (ITE) and provide theoretical supported uncertainty quantification for ITE estimates, improving personalization in marketing strategies.
- Addressed endogeneity in observational data using double machine learning, ensuring robustness in causal inferences.
- Validated methodologies by demonstrating convergence between observational and experimental results, reinforcing the effectiveness of conformal inference in complex models.
- Working Paper: Conformal Inference for Individualized Treatment Effects in Cross-Sectional CRM Data

Marketing Division, Research Assistant, Columbia University **Mar. 2024 – Aug. 2026**
Deep Learning, Machine Learning, NLP, AI, Multimodal Learning *with Profs. Hortense Fong and George Gui*

- Demonstrated the relationship between consumers' behavior and sentiment features extracted from book stories and movie trailers.
- Conducted sentiment analysis on over 800,000 book stories and movie synopses using fine-tuned GPT, RoBERTa, and VADER models.
- Performed psychological theme analysis with guided Latent Dirichlet Allocation (LDA) on extensive literary datasets.
- Implemented multimodal learning techniques on movie trailers by integrating music and text analysis for time-varying sentiment evaluation.

Services

Math GSI: Two semesters for Advanced Linear Algebra
Web page maintainer: Maintain the web page for lab for 2 years

Honors & Awards

- 04/30/2021 University Honors
- 12/20/2021 University Honors
- 12/19/2022 University Honors
- 03/20/2022 James B. Angell Scholar
- 03/19/2023 James B. Angell Scholar